

the skills to build a full operating system from code to a running device. How you progress from there is up to you – maybe you’ll add a feature, fix a bug, add a translation, or use what you’ve learned to build a new app or port to a new device.

What you’ll need

- A OnePlus 5
- A relatively recent 64-bit computer (Linux, OS X, or Windows) with a reasonable amount of RAM and about 100 GB of free storage (more if you enable ccache or build for multiple devices). The less RAM you have, the longer the build will take (aim for 8 GB or more). Using SSDs results in considerably faster build times than traditional hard drives.
- A USB cable compatible with the OnePlus 5
- A decent internet connection and reliable electricity.
- Some familiarity with basic Android operation and terminology. It would help if you’ve installed custom roms on other devices and are familiar with recovery. It may also be useful to know some basic command line concepts such as `cd`, which stands for “change directory”, the concept of directory hierarchies, and that in Linux they are separated by `/`, etc.

TIP

If you are not accustomed to using Linux, this is an excellent chance to learn. It’s free – just download and run a virtual machine (VM) such as VirtualBox, then install a Linux distribution such as Ubuntu (AOSP vets Ubuntu as well). Any recent 64-bit version should work great, but the latest Long Term Support (LTS) version is recommended. There are plenty of instructions on setting up VirtualBox to run Ubuntu, so we’ll leave that to you.

Build LineageOS and LineageOS Recovery Install the SDK

If you haven’t previously installed `adb` and `fastboot`, you can download them from Google. Extract it running:

- `unzip platform-tools-latest-linux.zip -d ~`

Now you have to add `adb` and `fastboot` to your `PATH`. Open `~/.profile` and add the following:

- `# add Android SDK platform tools to path`